

LOGISTIC REGRESSION AND DISCRIMINANT ANALYSIS COMPARAISON

Data Set: DEPRES.DAT

DATA: DEPRES.DAT

According to the World Health Organization, depression is the most prevalent mental disorder in the World. In the United States, 8.3% people are suffering from depression. This is the second highest percentage in the world only behind Brazil at 10%. **DEPRES.DAT** gives information about a depression study on 294 adults living in Los Angeles County. The objective of the study is to review both the incidence and prevalence of depression. And finally, it should determinate causality and consequences of having the condition for the respondents. There are 36 categorical variables in this data set. The other 2 variables are age and income for a total of 38 variables to analyze the condition. Variable 1 to 9 variable provides information about the respondents' identification and background. Variable 10 to 29 provides survey opinions of the respondents' feelings as it related to standard descriptions of depression scale items. Variable 30 provides a value in the depression index for each respondent. The Center for Epidemiologic Studies Depression Scale (CESD) is the tool used to quantify depression in the study. The threshold is 16 to assess if the respondent has depression (above 16) or not (below 16). CESD is a unique variable. The threshold of 16 is indeed based on C1 through C20. Each variable is scored on a 0-3 scale. 0 represents less than 1 day, 1-2 days is 1, 3-4 days is 2 and 5-7 days has a value of 3. However, the threshold decision is unknown. It can be hypothesized that some variables are weighted heavier in the analysis based on medical research. Finally, variable 32 to variable 38 provides respondents health information. The goal is to find a model with all necessary variables to discriminate between two groups. One group that has the condition and the other group who do not.

For our analysis purpose, C1 through C20 and CESD will be removed, and Y will represent CASES (above threshold). By doing so, we will be able to predict similar outcome based on the chosen set of variables in our model with appropriate accuracy. This technique can be used to create a screening tool in the form of a survey based on the selected variables to raise awareness to the respondents if they need to be treated for the condition. Since some variables have different units and scales. We will have a throughout discussion on either or not to standardize the data to avoid one variable to overwhelm the others in our final model. Furthermore, we have intercorrelation between some variables. Indeed, multicollinearity should be considered to optimize the model and avoid redundancy in our final function. To conclude there not missing data in the data sets which is valuable for the model.

Procedure and interpretation

- **Discriminant Analysis on DEPRES.DAT**
- **Step 1: Explore the Data**

Process:

The data set was classified based on the depressed cases against the non-depressed cases. Indeed, the first 50 rows represents depressed cases, and the remaining 244 rows is associated with subjects determined to not have depression. Based on the data summary (**Fig.1**) we can discuss several interpretations below. Also, numerical variables such as age and income can skew our model therefore standardization may be considered. However, all other variables have been properly scales. Indeed, drink, treatment and sex have been changed from 1-2 to 0-1 to match with the other 0-1 variables. Furthermore, the variables education, employ, marital, religion and health have all been changed to a 7-point scale.

```
> ## Step 1 - Explore the data
> summary(data)
```

SEX	AGE	MARITAL	EDUCAT	EMPLOY	INCOME	RELIG	CESD
Min. :1.000	Min. :18.00	Min. :1.000	Min. :1.00	Min. :1.000	Min. : 2.00	Min. :1.000	Min. : 0.000
1st Qu.:1.000	1st Qu.:28.00	1st Qu.:2.000	1st Qu.:3.00	1st Qu.:1.000	1st Qu.: 9.00	1st Qu.:1.000	1st Qu.: 3.000
Median :2.000	Median :42.50	Median :2.000	Median :3.00	Median :1.000	Median :15.00	Median :1.000	Median : 7.000
Mean :1.622	Mean :44.41	Mean :2.374	Mean :3.48	Mean :2.109	Mean :20.57	Mean :1.983	Mean : 8.884
3rd Qu.:2.000	3rd Qu.:59.00	3rd Qu.:3.000	3rd Qu.:4.00	3rd Qu.:3.000	3rd Qu.:28.00	3rd Qu.:3.000	3rd Qu.:12.000
Max. :2.000	Max. :89.00	Max. :5.000	Max. :7.00	Max. :7.000	Max. :65.00	Max. :6.000	Max. :47.000
CASES	DRINK	HEALTH	REGDOC	TREAT	BEDDAYS	ACUTEILL	CHRONILL
Min. :0.0000	Min. :1.000	Min. :1.000	Min. :1.000	Min. :1.000	Min. :0.0000	Min. :0.0000	Min. :0.0000
1st Qu.:0.0000	1st Qu.:1.000	1st Qu.:1.000	1st Qu.:1.000	1st Qu.:1.000	1st Qu.:0.0000	1st Qu.:0.0000	1st Qu.:0.0000
Median :0.0000	Median :1.000	Median :2.000	Median :1.000	Median :1.000	Median :0.0000	Median :0.0000	Median :1.0000
Mean :0.1701	Mean :1.204	Mean :1.772	Mean :1.187	Mean :1.497	Mean :0.2143	Mean :0.2959	Mean :0.5068
3rd Qu.:0.0000	3rd Qu.:1.000	3rd Qu.:2.000	3rd Qu.:1.000	3rd Qu.:2.000	3rd Qu.:0.0000	3rd Qu.:1.0000	3rd Qu.:1.0000
Max. :1.0000	Max. :2.000	Max. :4.000	Max. :2.000	Max. :2.000	Max. :1.0000	Max. :1.0000	Max. :1.0000

Figure 1: Data Summary

Interpretation:

As it was the case in the case study on chronic kidney disease, ID should not be considered in our final function. Indeed, ID is associated with each subject or observation therefore these values will be disregarded in our analysis on both methods. Furthermore, there are not any missing data. So, techniques such as mean imputation are not required. This aspect is appealing to our final model since there would be no errors associated with a statistical technique to replace or even omit missing data. Finally, $\sum(C1 \text{ to } C20) = \text{CESD}$ with a threshold at 16. If a subject has a value above 16 counts, the answer is true(1=depressed). If this condition does not hold therefore the answer is false (0=not depressed). Due to the collinearity between variables C1 to C20 to the depression index; we will omit C1 to C20 values and CESD in our model.

- Step 2: t.test

Process:

T-test is used to determine if there is a significant difference between the mean of two groups i.e., is the subject depressed or not? In other words, t-test determine separability if variable selection is important. It should be noted that t-test assumes that both groups are independent. Also, it assumes that both variances are equal. We will use the function `t.test` in R while considering both assumptions in our interpretation.

Interpretation:

We ran t-tests as shown in **Fig.2** to verify the relevancy of our model. The threshold to keep or drop variables is $p=0.5$. It should be noted that variable age is dropped which will become validated by our proposed model in our logistic regression below. We end up with 5 significant variables as shown in the summary in **Fig 3**. Beddays and income have the highest variance in xbars. The correlation between income and beddays as it related to depression is interesting to review. For a respondent without depression, the variance in income is higher than a respondent that has depression. Similarly, the variance in beddays is smaller when a subject does not have a condition compared to a respondent that does have it. Depression is such a complex condition that some depressed people may not sleep much. But our model shows that depressed people would sleep more. Also, an individual with a decent job has less chance to be depressed than another one with a high paying job with a lot of stress. Without a physical exam and/or a thyroid check, it is complicated to determine if you have depression or not (Mayo Clinic). However, it is redhibitory to categorize the cause of depression only based on income, beddays alone. Therefore, additional variable such as gender(sex) become necessary asset to our discrimination model. It should be noted there is a higher variance in variable sex for respondent with the condition. According to the Los Angeles LGBT Center website, several residents are not open about their preferred gender based on social peer pressure. This can be a cause of depression for a subset of respondents.

- **Step 3: Summary Characteristics**

Process:

The objective of this process is to highlight the difference between the mean values for each variable in the data set based on the final answer about depression ((true=CASES or false). The goal is to determinate discriminator variables. These variables give the best discrimination between the two groups. Therefore, we computed `xbar=colmean` in R to get **Fig. 4** below.

```
> xbar1
  SEX  EMPLOY  INCOME  HEALTH  BEDDAYS
1.80   0.52   15.20   2.06   0.42
> xbar2
  SEX      EMPLOY      INCOME      HEALTH      BEDDAYS
1.5860656  0.2540984 21.6762295  1.7131148  0.1721311
```

Figure 4: Variables:sex, employ, income, health and beddays values for xbar1 and xbar 2

Interpretation:

Opposite to our data set, there is not any correlation between education and depression according to our research (LA Health). Intuitively, it can be argued that somebody with low education has a decent chance to experiment depression. We can make this assumption since higher education is correlated to higher income and better lifestyle. However, it should be also noted that that wealthier nations which are associated with high percentage of people with high education; have higher percentage of depression as well (PulseTMS).

As far as the data is concerned, we will have a threshold a 5 for education variable. If you are in the categories 5 to 7, respondent is classified as 1. If they have less than 5, they are classified as zero. Furthermore, full time workers and retired workers would be classified as 0 and all other options as 1.

There is not a relevant difference between the level of depression between religious and non-religious subjects. A study by the neuroscience news shows that religion can stave off depression for teenagers (sample size =12000 teens). However, it is difficult to get religious information for both transparency and privacy reasons. Furthermore, marital status is not a good indicator of depression which is logical from an intuitive standpoint. Indeed, it does not matter if you are married or not, subjects can experience depression.

Contrary to religion and marital status, there is a clear difference in depression between people that estimate their health as excellent and good (1-2) versus fair and poor (3-4). Therefore 2

would represents respondents with fair or poor health. And 1 would represent respondents with excellent and good health in our processed data set.

	True Condition	False Condition		True Condition	False Condition
BEDDAYS	0.42	0.17	BEDDAYS	0.24	0.14
EMPLOY	0.52	0.26	EMPLOY	0.26	0.19
INCOME	15.2	21.67	INCOME	96.77	255.21
<i>The condition is being depressed</i>			<i>The condition is being depressed</i>		

Table 1A,1B:Variance&Co-variance

As shown on **Fig.4**, there is a relevant difference in means for three variables: beddays, employ and income for respondents in each group. If the difference is high enough therefore the variable would be part of our model based on discriminant analysis. As discussed earlier higher income may be connected to non-depressed respondents. Furthermore, people who have a job or retired tends to be more depressed than does who do not. Both variance and co-variances indicates significant differences between being depressed or not as shown on **Table 1A&1B**.

- **Step 4: Calculate Linear Combination and Weight**

Process:

The scores for the 294 respondents are calculated by matrix multiplying the difference between the means and the transpose of the original data multiplied by the pooled variances). The weights for the variables are calculated by matrix multiplying the transpose of the difference in means and pooled variance matrix (**Fig.5**). Astar represents the coefficients of the variable for the discriminant analysis. Each coefficient is correlated to the respective weight of each variable in your final function. The weights are set up to maximize the separation between the two groups and their sum equal to 1(**Fig.6**).

```
> round(astar,2)
      [,1]
SEX      0.18
EMPLOY   0.34
INCOME   0.01
HEALTH   0.10
BEDDAYS  0.36
```

Figure 6: Coefficients ranking

Interpretation:

As shown in **Fig.6**, beddays and employment have the most impact to differentiate both groups at 36% and 34% respectively. We see that the income is in fifth place at 1% according to astar output. Therefore, the initial assumption in which depression is highly correlated to income is questionable according to our model.

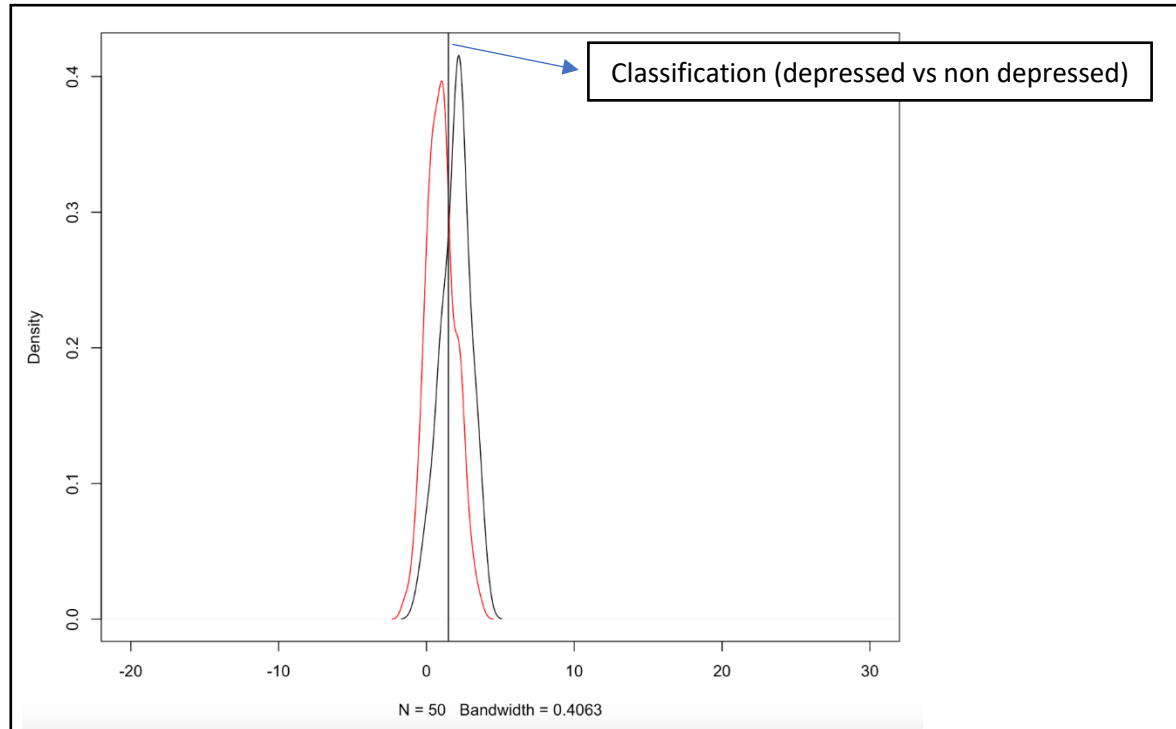


Figure 7: variance is similar between both group which is in line with t test assumption. However, means are different between the two groups. Looking at the vertical distance from the probability density, the ratio is inversed on both side of the cutoff. In other words, red curve is higher than blue curve before cutoff and vice versa. The black line represents depressed subjects and the red line non depressed subjects.

- **Step 5 through 8: Calculate cutoff and classify, inspect results and calculate separation, calculate upperbound and separation and calculate accuracy metrics**

Process:

The R inputs used to calculate cutoff, separation, upperbound and accuracy metrics are shown in **Fig.8**. The cutoff is set at an adequate value as we will not lose much accuracy than before based on the data set. We still do have misclassifications which is expected but in terms of true positives for the condition; our threshold is appropriate.

```

> cutoff
[1] 1.486521
> ## Step 7 - Calculate upperbound and separation
> upper=(xbar1-xbar2)%*%solve(Sp)%*%(xbar1-xbar2))
> upper
      [,1]
[1,] 0.9613685
> sy=(sum((y[pi1]-mean(y[pi1]))^2)+sum((y[pi2]-mean(y[pi2]))^2))/(length(pi1)+length(pi2)-2)
> separation=abs(mean(y[pi1])-mean(y[pi2]))/sy
> sy
[1] 0.9527611
> separation
[1] 1.009034
> t(a)%*%Sp%*(a)
      [,1]
[1,] 0.9613685

```

Figure 8: R input and output calculations for the cutoff, upperbound, separation and other metrics in our discriminant analysis

Interpretation:

The cut-off is calculated via the mean of the means scores between the two groups with the prior probabilities set as 0.5. It refers to the threshold for differentiating respondents between having depression and not having depression. As shown in **Fig.8**, the cutoff is 1.49.

The goal in creating Y is to make it so that the average values depressed vs non depressed with respect to y are the biggest apart. And, at the contrary we want the variance to be small. By doing so, we will optimize the separation between the two groups. Since our separation is 1. We can fairly be satisfied with our separation.

To conclude, we ran a linear discriminant analysis on the pre-processed data i.e.C1 through C20 variables as well as CESD variable were dropped from the model (**Fig.9**). Beddays and Sex are indeed significant which is not the case for income. This might be the reason why our separation is not bigger. Overall, our model is sound, but it is difficult shows to design a powerful model. While Factor analysis does not assume standard distribution like discriminant analysis, information on depression is not as necessary prior to assign predictive value. However, the strength of our model for our discriminant analysis was related to our empirical research on the condition.

Procedure and Analysis

- **Logistic Regression Analysis on DEPRES.DAT**
- **Step 1 through 3: Explore Data, Run the logistic regression with one variable& Run the logistic regression on all data and finally explore backward elimination**

Process:

The objective of logistic regression is to predict the condition i.e., being depressed or not using the necessary variables. It should be noted that some dropped variables might be relevant to better identify subjects that are depressed or not for health professionals. However, the goal is to create a model with reduced variables in R that would give a similar (or close) result about the respondents if all the variables were considered. For Steps 2&3, we use glm function in R. This function supports non normal distribution and efficient with a variable with non-constant variance(**educBA**). These two reasons justify why the function is appropriate for our data set to run a regression model.

Interpretation:

We reviewed the relationship between the selected variables. All independent variables are uncorrelated to each other. Therefore, our model will not be depreciated due to multicollinearity. Based on our prior assumption that income can be a predictor of depression. We ran a regression on CASES and INCOME which shows a p-value that is negatively correlated while being less than 0.5(**Fig.10**). Then, we ran another regression on the entire data set (**Fig.11**). Sex, Income, Religion and Beddays are the most correlated to CASES based on their p-values. It should be noted that, the graphic shows a lower AIC than the previous regression. Therefore, a backwards regression is required to complete the model (**Fig.12**). The remaining variables are Sex, Age, Education, Income, Religion, Health and Beddays. Health and Education could be dropped in our final model since they are insignificant. However, depression is associated with other illnesses in some cases. Indeed, according to LA Health, depressive disorders occur with other illness such as heart disease and diabetes so we will keep variable health while removing the variable education. This is a sound decision that might affect our overall model as the decision is made while disregarding the high p-value of health variable compared to the other five variables. Finally, it should be noted that education and income are negatively associated with depression. As discussed earlier via PulseTMS findings, there is indeed a high median income and overall higher education per people in wealthier countries. However, there is also a higher percentage of

depressed individuals. In the same way, the older people are the less likely they may be depressed. We can hypothesize that both social and professional expectations fade with age therefore we expected the negative correlation between age and CASES as well. A final regression was completed based on proposed model based on our research on depression (**Fig.13**). It would be interesting to compare the model based on literacy review to the model based on statistical review.

```
Call:
glm(formula = CASES ~ INCOME, family = "binomial", data = data)
> summary(model1)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-0.7884 -0.7012 -0.5631 -0.3878  2.2913

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -0.93757    0.26584  -3.527 0.000421 ***
INCOME      -0.03583    0.01348  -2.658 0.007863 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 268.12  on 293  degrees of freedom
Residual deviance: 259.40  on 292  degrees of freedom
AIC: 263.4

Number of Fisher Scoring iterations: 5
```

```
Call:
glm(formula = CASES ~ ., family = "binomial", data = data)
> summary(model1)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.6482 -0.6003 -0.3967 -0.2188  2.6631

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.801860    1.092163  -1.650  0.0990 .
1..SEX       0.903104    0.434343   2.079  0.0376 *
AGE         -0.026165    0.013418  -1.950  0.0512 .
MARITAL      0.018573    0.113981   0.163  0.8706 .
EDUCAT      -0.164578    0.167563  -0.982  0.3260 .
EMPLOY       0.138171    0.125107   1.104  0.2694 .
INCOME      -0.030625    0.015229  -2.011  0.0443 *
RELIGION     0.261389    0.119051   2.196  0.0281 *
DRINK       -0.566399    0.487303  -1.162  0.2451 .
HEALTH       0.208460    0.133640   1.560  0.1188 .
REGDOC       0.306071    0.439135   0.697  0.4858 .
TREAT       -0.325237    0.427208  -0.761  0.4465 .
BEDDAYS      0.942031    0.431214   2.185  0.0289 *
ACUTEILL     0.064955    0.417490   0.156  0.8764 .
CHRONILL     0.004826    0.427503   0.011  0.9910 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 268.12  on 293  degrees of freedom
Residual deviance: 222.50  on 279  degrees of freedom
AIC: 252.5

Number of Fisher Scoring iterations: 5
```

Figure 10 & 11: Regression CASES ~ INCOME (one variable) AND Regression CASES~. (preprocessed data).

```

> summary(model3)

Call:
glm(formula = CASES ~ i..SEX + AGE + EDUCAT + INCOME + RELIGION +
    HEALTH + BEDDAYS, family = "binomial", data = data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.5466  -0.6142  -0.4087  -0.2261   2.8957

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -1.68960    0.93456  -1.808  0.0706 .
i..SEX       0.97422    0.41452   2.350  0.0188 *
AGE         -0.02169    0.01026  -2.113  0.0346 *
EDUCAT      -0.21826    0.15704  -1.390  0.1646
INCOME      -0.03209    0.01458  -2.202  0.0277 *
RELIGION     0.28589    0.11422   2.503  0.0123 *
HEALTH       0.19568    0.12141   1.612  0.1070
BEDDAYS      1.06659    0.38085   2.801  0.0051 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 268.12  on 293  degrees of freedom
Residual deviance: 225.63  on 286  degrees of freedom
AIC: 241.63

Number of Fisher Scoring iterations: 5

Call:
glm(formula = CASES ~ ., family = "binomial", data = data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.4055  -0.5863  -0.4318  -0.2800   2.6801

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept) -2.99743    0.86204  -3.477 0.000507 ***
SEX          0.55628    0.40500   1.374 0.169590
EMPLOY       0.97151    0.34243   2.837 0.004552 **
INCOME      -0.03442    0.01390  -2.477 0.013258 *
HEALTH       0.21312    0.18938   1.125 0.260434
BEDDAYS      1.10562    0.36326   3.044 0.002338 **
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 268.12  on 293  degrees of freedom
Residual deviance: 231.02  on 288  degrees of freedom
AIC: 243.02

Number of Fisher Scoring iterations: 5

```

Figure 12&13: Backward Elimination Exploration AND Proposed Model based on Research

- **Step 4 and 5: Explore your new model, conduct Hypothesis test on the model /Alternate. Ho: Model Fits the Data, Ha: Model does not Fit, Definition 5-2. Then compare 2 models.**

Process:

We will use the `confint()` function in R to explore our models. This function computes confidence intervals for one or more parameters in a fitted model. Our outputs are **Fig.14&Fig. 15** that would be interpreted below.

```

> confint.default(model3)

                2.5 %      97.5 %
(Intercept) -3.52130953  0.142105859
i..SEX       0.16177907  1.786652106
AGE         -0.04180528  -0.001572968
EDUCAT      -0.52605112  0.089521261
INCOME      -0.06066234  -0.003521690
RELIGION     0.06201683  0.509761840
HEALTH      -0.04227973  0.433630001
BEDDAYS      0.32014571  1.813038314

```

Figure 14&15: Confidence Interval for True model and Proposed Model Respectively

Interpretation:

In our statistical model, age, sex, income, religion and beddays are relevant variables and therefore the model is not perfect since it does not match with our prior p-values results. Furthermore, religion and health are indeed insignificant to the model as both variables have 0

within the confidence interval. Therefore, our assumption to remove the variable education was a good one since the resulting model while keeping that variable (as well as health) is weak. On the other hand, our proposed model is stronger. Variable age and religion are not included as our research showed a lack of significance of these variables as it related to depression. However, beedays, sex, health, employ becomes relevant in our proposed model which is more in line with our p-values results. Also, the 95% CI for the selected variables does not pass through 0 except for all selected variables and thus are validated in our proposed model. Then, several hypothesis tests were run to determine the strength of the models which showed the null deviance and residual deviance returning a low p-value justifying that both models are acceptable (**Fig.16&Fig.17**). Furthermore, to verify the model quality, we also compare null deviance to residual deviance. Null deviance is 37.10 more than residual deviance (proposed model). This is also true for the pure statistical model with null deviance equal to 42.49. And we computed the p-value of a chi-square test that compares the selected model to the null model. The p-value is 5.697531e-07. The value is lower than the standard threshold of 0.05. The above results confirms that the selected model is better than the null model. This statement is also true for the proposed model using chi square test. Indeed, as shown in **Fig. 17**, we can conduct a chi-square test to compare the deviance and the residuals.

H_0 : The model fits the data.

H_a : The model does not fit the data.

The p-value is closed to 1 which implies that the null hypothesis is true therefore the model fits the data.

```
> with(model3, null.deviance - deviance)
[1] 42.49822
> ##df
> with(model3, df.null - df.residual)
[1] 7
> ## pvalue of difference
> with(model3, pchisq(null.deviance - deviance, df.null - df.residual, lower.tail = FALSE))
[1] 4.168701e-07
> |
```

Figure 16: Hypothesis Test-Statistical Model

```

> with(model, null.deviance - deviance)
[1] 37.10825
> ##df
> with(model, df.null - df.residual)
[1] 5
> ## pvalue of difference
> with(model, pchisq(null.deviance - deviance, df.null - df.residual, lower.tail = FALSE))
[1] 5.697531e-07
~
> with(model, pchisq(deviance, df.residual, lower.tail = FALSE))
[1] 0.9941928

```

Figure 17: Hypothesis Test-Proposed Model

- **Step 6 through 8: Predict probabilities and Odds Ratios of New Data, predict and Plot in Sample Data**

Interpretation:

As shown in **Fig.18**, the model can predict if a respondent has the condition or not with a decent accuracy. However, the data has a large range therefore both medical and empirical research should be done to determinate the adequate threshold to avoid misclassification. Therefore, we will look at the output with two different thresholds at 0.5 and 0.25 respectively as shown in **Fig.19**. The threshold of .5 provides a perfect true positive rate and a low rate of false positives. However, it gives a significant negative rate of 16%. This is a relatively high cost of misclassification. As far as the .25 threshold is concerned, there is an increase in f measure but a decrease in accuracy. Indeed, the true positive rate is not perfect with this constraint and there is a sharp increase in both false positive rate and false negative rate, respectively. This is not optimal.

However, we can graphically optimize our model to minimize the cost of misclassification. The ROC curve (**Fig.20**) shows that sensitivity at 83% is ideal because it gives an optimized result of true and false negative. Therefore, the ideal threshold should be in between 0.5 and 0.25. This model is going to be able to predict if a respondent has depression based on sex, age, income, religion, health education and beddays. However, the proposed model would make stronger decision as the Area under the ROC curve is greater than the statistical model (**Fig.20**). It shows that medical research based on empirical data gave a better model than a pure statistical analysis of our sample of 294 respondents. Therefore, exploring the data becomes the most critical step in statistical analysis. Because one variable can be significant in a model based on p value or other

test but can be deemed insignificant based on literacy review. As a conclusion, we will use our proposed model with Sex, employ, beddays, income and health to make better decision on depression for prospective respondents (**Fig.21**).

```

Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.00567 0.06408 0.12576 0.17007 0.23052 0.81578
> round(summary(phat3), 2)
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.01   0.06   0.13   0.17   0.23   0.82

```

Figure 18: Prediction Statistical Model

```

> c_accuracy(data$CASES,classify) # to run this you must run my code below first.
recall precision accuracy tpr fpr fmeasure tp tn fp fn
0.10000000 0.45454545 0.82653061 0.10000000 0.02459016 0.16393443 5.00000000 238.00000000 6.00000000 45.00000000

> classify1=ifelse(phat3>.25,1,0)
> summary(classify1)
Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
0.0000 0.0000 0.0000 0.2211 0.0000 1.0000
> c_accuracy(data$CASES, classify1)
recall precision accuracy tpr fpr fmeasure tp tn fp fn
0.60000000 0.4615385 0.8129252 0.60000000 0.1434426 0.5217391 30.00000000 209.00000000 35.00000000 20.00000000

```

Figure 19: Classifications at 0.5 threshold and 0.25 threshold, respectively.

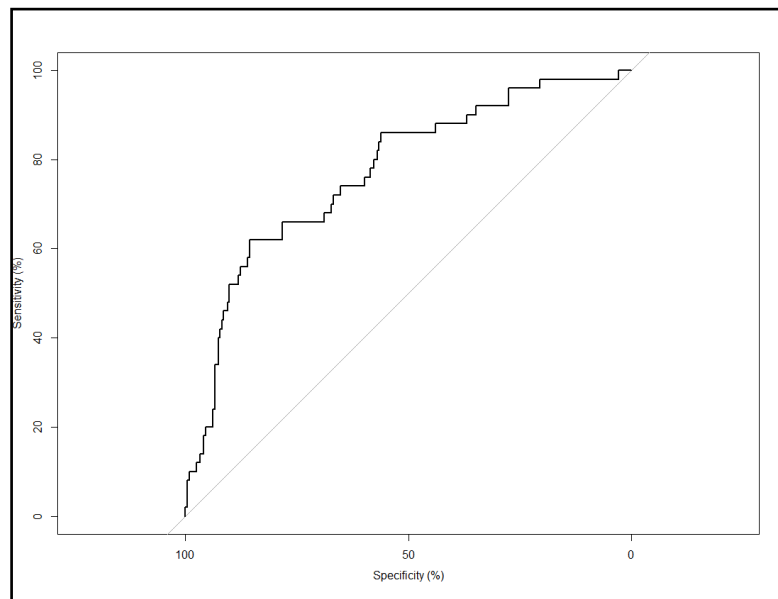


Figure 20: ROC CURVE

- **Conclusion and Recommendation**

Discriminant analysis approaches the problem in a different way than logistic regression. It does classification in a single step and it does not give probabilities and coefficients. However, it does give weightings to determinate the necessary variable to separate between a subject with depression or not. Contrary to PCA where the goal is to determinate the new variable with the most variances; discriminant analysis has an objective to create new values such that condition is true on one end and condition is false on the other end. Indeed, you are trying to maximize the separation while PCA is trying to maximize variance. Therefore, discriminant analysis is truly different than PCA even though there are similarities in both methods.

As for the comparison between discriminant analysis and logistic regression, the latter focus on probabilities and the other on decision making. Based on the probabilities, you can create a regression to minimize misclassification. For certain disease where the cost of misclassification is high; I would use logistic regression. Indeed, there are two costs of misclassification. If we guess that they are not depressed but this is not the case as the subject has the condition, this is a false negative (C2 given 1)). Or if they are not depressed and we guess that they did, this is a false positive (C1 given 2)). Both false positive and false negative can be respectively calculated via the integral from cutoff (lower bound) to infinity and infinity to the cutoff (upper bound) of the function associated with the black curve. The goal is to minimize those costs especially the cost associated with false negative. Hence my recommendation for logistic regression for this specific condition.

To conclude, logistic regression designs a model based on the conditional distribution of the response given a set of variables. Discriminant analysis models the distribution of the predictors separately for each response class and then uses *Bayes theorem* to convert these into estimates. If the distributions are normal, the model will be very similar to logistic regression. Linear Discriminant Analysis works better than logistic regression when the two classes of the response are well-separated which was arguably not the case for **DEPRES.DAT**.

References

- PulseTMS: Depression Guide: Los Angeles
<https://pulsetms.com/resources/depression-los-angeles/>
- Center for Epidemiologic Studies Depression Scale (CES-D), NIMH
[Center for Epidemiologic Studies Depression Scale \(CES-D\) \(brown.edu\)](#)
- Mayo Clinic: Depression (major depressive disorder)
[Depression \(major depressive disorder\) - Symptoms and causes - Mayo Clinic](#)
- Study examines religious experience and depression
[Study Examines Religious Experiences and Depression - Neuroscience News](#)
- LA Health: Depressive disorders Among Los Angeles County Adults (2002)
[LA Health Depression \(lacounty.gov\)](#)
- Learn how to construct a general model in R
<https://www.educba.com/glm-in-r/>
- Los Angeles LGBT Centers
[transgender health program - Los Angeles LGBT Center \(lalgbtcenter.org\)](#)

Appendix

```
> ## Step 2 - t.test
> t.test(data$AGE[1:244],data$AGE[245:294])

Welch Two Sample t-test

data: data$AGE[1:244] and data$AGE[245:294]
t = 1.7866, df = 72.548, p-value = 0.07817
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.5621317 10.2857383
sample estimates:
mean of x mean of y
 45.2418  40.3800

> t.test(data$INCOME[1:244],data$INCOME[245:294])

Welch Two Sample t-test

data: data$INCOME[1:244] and data$INCOME[245:294]
t = 3.7507, df = 109.8, p-value = 0.0002834
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 3.054257 9.898202
sample estimates:
mean of x mean of y
21.67623 15.20000

> t.test(data$BEDDAYS[1:244],data$BEDDAYS[245:294])

Welch Two Sample t-test

data: data$BEDDAYS[1:244] and data$BEDDAYS[245:294]
t = -3.3248, df = 61.071, p-value = 0.0015
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.39693938 -0.09879833
sample estimates:
mean of x mean of y
0.1721311 0.4200000

> t.test(data$ï..SEX[1:244],data$ï..SEX[245:294])

Welch Two Sample t-test|

data: data$ï..SEX[1:244] and data$ï..SEX[245:294]
t = -3.2764, df = 81.997, p-value = 0.001543
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.34382999 -0.08403886
sample estimates:
mean of x mean of y
0.5860656 0.8000000
```

```

> t.test(data$.data_1.166666667[1:244],data$.data_1.166666667[245:294])

Welch Two Sample t-test

data: data$.data_1.166666667[1:244] and data$.data_1.166666667[245:294]
t = 1.6691, df = 70.545, p-value = 0.09954
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.02516198  0.28352264
sample estimates:
mean of x mean of y |
0.5491803 0.4200000

> t.test(data$.data_2.333333333[1:244],data$.data_2.333333333[245:294])

Welch Two Sample t-test

data: data$.data_2.333333333[1:244] and data$.data_2.333333333[245:294]
t = 0.28081, df = 71.981, p-value = 0.7797
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.09898346  0.13144248
sample estimates:
mean of x mean of y
0.1762295 0.1600000

> t.test(data$.data_3.5[1:244],data$.data_3.5[245:294])

Welch Two Sample t-test

data: data$.data_3.5[1:244] and data$.data_3.5[245:294]
t = -0.86305, df = 63.752, p-value = 0.3913
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.15161566  0.06014025
sample estimates:
mean of x mean of y
0.0942623 0.1400000

> t.test(data$.data_4.666666667[1:244],data$.data_4.666666667[245:294])

Welch Two Sample t-test

data: data$.data_4.666666667[1:244] and data$.data_4.666666667[245:294]
t = -1.2455, df = 64.743, p-value = 0.2174
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.21810872  0.05056774
sample estimates:
mean of x mean of y
0.1762295 0.2600000

> t.test(data$.data_7[1:244],data$.data_7[245:294])

Welch Two Sample t-test

data: data$.data_7[1:244] and data$.data_7[245:294]
t = -0.7789, df = 53.183, p-value = 0.4395
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 -0.05684686  0.02504358
sample estimates:
mean of x mean of y
0.004098361 0.020000000

```

Figure 2: T-tests

```

> Y
[1] -0.63871936 0.51358550 0.18301743 -0.46752420 -0.80761030 0.37662937 -0.39904613 0.30815130 1.19836616 -1.18423966
[11] 0.95869293 -0.33056806 -0.26209000 -1.35543483 -1.49239096 0.30815130 -0.57024130 0.42269067 -0.53600226 -0.12513387
[21] -0.57024130 -0.12513387 -0.33056806 0.17119517 -0.09089483 0.65054163 -0.46521996 -0.03193480 0.01182227 0.01182227
[31] 0.51358550 -0.12513387 0.78749777 0.78749777 0.37662937 0.78749777 -0.12282963 0.95869293 0.01182227 -0.46752420
[41] 1.19836616 1.10747133 -1.62934709 0.17119517 0.18301743 -0.60448033 0.23967323 -0.11822227 -0.57024130 -0.09089483
[51] -0.57024130 1.19836616 0.78749777 -0.95638870 -0.46752420 0.18301743 -0.39904613 -1.86902033 0.76508100 -0.33056806
[61] -0.36480710 0.99523620 0.23967323 0.37662937 -1.83478129 -0.63871936 -0.53600226 -0.33056806 1.19836616 1.19836616
[71] -1.62934709 -1.56086903 0.95869293 -0.12513387 0.06847807 0.27391227 -0.26209000 -0.12513387 1.44986167 -0.53600226
[81] -1.73206419 0.13695613 1.54075650 -0.12513387 -0.50176323 2.22553716 -0.03193480 -1.76630323 -1.38967386 0.78749777
[91] -0.30584707 -0.44280320 -0.12513387 -1.49239096 -0.50176323 -0.46752420 0.99523620 -0.46752420 0.65054163 -0.53600226
[101] 0.30815130 -0.71671546 0.51358550 0.23967323 -0.50176323 -0.33056806 1.19836616 0.51358550 0.76508100 1.54075650
[111] -1.69782516 1.54075650 0.95869293 1.19836616 -0.46752420 0.65054163 0.95869293 0.44510743 -1.83478129 -1.73206419
[121] -0.60448033 -0.39904613 -0.12513387 0.54782453 0.42269067 0.31997357 -0.80761030 2.22553716 0.66236390 -0.60448033
[131] 0.45692970 0.99523620 1.54075650 1.10747133 -0.12513387 -0.50176323 0.65054163 -0.46752420 0.17119517 -0.57024130
[141] 0.13695613 1.10747133 0.78749777 0.78749777 0.51358550 0.30815130 0.65054163 -0.57024130 0.78749777 -0.60448033
[151] -1.49239096 -0.46752420 -0.70719743 0.76508100 0.13695613 -0.40856417 0.76508100 -0.46752420 -0.34008610 0.01182227
[161] -0.53600226 -0.12513387 0.65054163 -0.57024130 -0.50176323 0.95869293 0.95869293 0.01182227 -1.21847870 0.18301743
[171] 0.20543420 0.27391227 0.17119517 -1.49239096 -0.39904613 -0.12513387 0.42269067 -0.39904613 0.76508100 1.44986167
[181] 0.21956070 1.10747133 0.44510743 0.28573453 -1.76630323 -0.26209000 -0.50176323 1.88314683 1.54075650 0.42269067
[191] 1.44986167 1.88314683 -0.53600226 -0.33056806 0.65054163 -0.46521996 -0.46521996 -0.73913223 0.01182227 -0.12513387
[201] -0.53600226 -0.71671546 -0.26209000 -0.46752420 0.18301743 -0.50176323 -1.69782516 0.30815130 0.65284587 -0.26209000
[211] -0.78519353 1.19836616 -0.26209000 0.44510743 -0.33056806 -0.46752420 0.78749777 0.30815130 0.51358550 -0.26209000
[221] 0.01182227 2.22553716 0.42269067 0.44510743 0.51358550 -0.39904613 0.37662937 -0.50176323 -0.57024130 0.78749777
[231] -0.57024130 1.19836616 1.54075650 1.19836616 0.18301743 -0.63871936 0.44510743 0.42269067 1.19836616 0.78749777
[241] 1.19836616 1.88314683 0.95869293 -0.46752420 -1.21847870 0.08030033 -1.80054226 -1.73206419 -0.12513387 -1.49239096
[251] 0.42269067 0.01182227 -1.49239096 0.95869293 1.54075650 -0.57975933 -1.16182290 0.23967323 -1.59510806 -0.53600226
[261] 0.37662937 -0.80761030 -0.33056806 -1.69782516 -0.26209000 0.01182227 -0.70719743 -1.49239096 -1.93749839 -0.63871936
[271] -0.46752420 -1.35543483 -1.62934709 0.06847807 -0.33056806 -0.09089483 -1.62934709 -0.53600226 0.65054163 -0.57024130
[281] -0.33056806 -0.53600226 -1.76630323 -0.33056806 -1.35543483 -0.44280320 0.18301743 -0.39904613 -0.53600226 -1.93749839
[291] -0.39904613 -1.04728353 0.95869293 -1.04728353

> a
[,1]
1..SEX -0.77567550
INCOME 0.03423903
BEDDAYS -1.23030096
> astar
[,1]
1..SEX -0.38019293
INCOME 0.01678207
BEDDAYS -0.60302501

```

Figure 5: Combinations Matrix

Table 1. Percentage of Los Angeles County Adults (18 and Older) Ever Diagnosed with a Depressive Disorder, 1999

	Percentage	(±95% CI)	Estimate
<i>Los Angeles County</i>	9%	±1	623,000
<i>Gender</i>			
Male	7%	±1	230,000
Female	11%	±1	393,000
<i>Age Group</i>			
18 to 24	7%	±1	57,000
25 to 29	7%	±2	53,000
30 to 39	7%	±1	128,000
40 to 49	11%	±2	160,000
50 to 59	13%	±2	120,000
60 to 64	9%	±3	28,000
65 or over	8%	±2	78,000
<i>Race/Ethnicity</i>			
Latino	7%	±1	188,000
White	12%	±1	326,000
African-American	9%	±2	65,000
Asian/Pacific Islander	4%*	±1	39,000
<i>Federal Poverty Level¹</i>			
Less than 100% FPL	9%	±2	112,000
100% to < 200% FPL	11%	±1	176,000
200% to < 300% FPL	9%	±1	129,000
300% FPL or above	8%	±1	207,000
<i>Employment Status</i>			
Employed	7%	±1	302,000
Unemployed	13%	±4	28,000
Not in workforce ²	12%	±1	273,000

Table 2: Table from LA Health. In LA County, 7% male have ever been diagnosed with a depressive disorder while our sample size show that 14% respondents have the condition.


```

Call:
glm(formula = CASES ~ CESD, family = "binomial", data = data)

Deviance Residuals:
    Min       1Q   Median       3Q      Max
-1.198e-04 -2.100e-08 -2.100e-08 -2.100e-08  1.257e-04

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  -579.88   65436.59  -0.009    0.993
CESD           37.41    4225.40   0.009    0.993

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2.6812e+02  on 293  degrees of freedom
Residual deviance: 1.6515e-07  on 292  degrees of freedom
AIC: 4

Number of Fisher Scoring iterations: 25

```

Figure 22: Proposed model regression. As Cases is calculated only based on CESD, backward elimination should give us a one variable model using CESD. Running backward elimination confirms this hypothesis.